

Beyond Keywords: Attack Signatures and a Diagnostic Benchmark for Multilingual Scam Detection in Indian Languages

Ram Nivas Attanti

Department of Computer Science and Business Systems
JAIN (Deemed-to-be University), Bengaluru, India
ramnivasattanti2008@gmail.com

September 20, 2026

Abstract

Scam-message classifiers are typically evaluated on held-out samples drawn from the same distribution as their training data, where high scores are easy to obtain and reveal little about whether a model has learned the structure of an attack or merely its vocabulary. We argue that the right unit of analysis is the *Attack Signature*: a typed tuple recording who the sender is impersonating, what pretext is deployed, which asset is sought, what the recipient is asked to do, and which persuasion levers are pulled. A signature should be invariant to surface realisation—the same fraud written in English, Hindi, Telugu, Tamil, Kannada, romanised transliteration or code-mixed script is one attack—and sensitive to security semantics, so that inverting a single clause (“*share your OTP*” → “*never share your OTP*”) changes it. We formalise this notion, define four diagnostic measures over it, and release BHARAT-SCAM-X, a constructed probe corpus of 1,823 labelled messages spanning nine Indian language varieties, twelve attack families and five evaluation conditions, including 143 minimal-edit contrast pairs whose members share 55.2% of their tokens but carry opposite security meaning. Evaluating four baselines, we find the gap we predicted. A character n -gram classifier reaches 0.942 macro- F_1 in-distribution but only 0.700 on semantic flips, and marks 50.7% of genuine security advisories as attacks. A structured model that predicts the signature and derives risk from it is substantially more flip-sensitive (0.559 vs. 0.175 flip-success) and never fires on our hard negatives, yet it recovers the correct signature core for only 5.6% of recombined attacks that use exclusively training-seen components. We conclude that in-distribution accuracy is close to uninformative for this task, and release the corpus, schema and harness to make the harder questions measurable.¹

1 Introduction

In calendar year 2025 the Indian Ministry of Home Affairs recorded roughly 28.15 lakh cyber-crime complaints and about Rs. 22,495 crore in reported financial losses [12]. A large share of this volume begins with a text message: an SMS or messaging-app note claiming that an account is about to be blocked, a parcel is held at customs, or a legal case has been filed, and asking the recipient to share a one-time password, install a screen-sharing application, or transfer money.

The natural engineering response is a classifier. The natural evaluation is a held-out split. Both are reasonable and both, we argue, are misleading. A model trained on messages containing *OTP*, *KYC*, *blocked* and *urgent* can score very highly on a held-out sample containing the same words without having represented anything about how the fraud works. Two failure modes follow directly, and neither is visible in a standard evaluation.

¹Code and data: see Section 11.

First, **surface variation across languages**. India’s messaging traffic is multilingual, frequently transliterated into Latin script, and often code-mixed. The messages “*Your bank account will be blocked. Share OTP.*” and “*Mee bank account block aipothundi. OTP cheppandi.*” are the same attack. They share almost no character n -grams.

Second, **security-semantic inversion**. The messages “*Your KYC has expired. Share the OTP sent to your phone.*” and “*Your KYC has expired. Never share the OTP sent to your phone with anyone.*” differ by one clause and share most of their tokens. The first is credential theft; the second is the advisory a bank sends to prevent it. A model keyed on vocabulary cannot separate them, and a deployed system that flags the second erodes exactly the trust that makes advisories work.

Contributions.

1. **The Attack Signature** (Section 3): a typed eight-field representation of a message’s underlying attack, with an agreement measure σ and a transparent risk map ρ .
2. **The BHARAT-SCAM-X probe corpus** (Section 4): 1,823 labelled messages across nine Indian language varieties and twelve attack families, generated by a compositional grammar so that every item carries a gold signature and the splits can be controlled exactly.
3. **Four diagnostics** (Section 5): cross-lingual signature consistency, semantic-flip sensitivity, compositional signature recovery, and open-set behaviour.
4. **A baseline study** (Section 7) quantifying the gap between in-distribution accuracy and every diagnostic, and showing that structured signature prediction closes part but not most of it.

What this paper does not claim. We want to be precise about the epistemic status of our results, because it bounds what can be concluded from them. BHARAT-SCAM-X is a *constructed probe corpus*, not a sample of field traffic. Its messages are composed from a template grammar, in the tradition of SCAN [11], COGS [10] and CheckList [13]. Accordingly, the absolute numbers in Section 7 are *not* estimates of deployment performance, and a model that scores well here is not thereby shown to be deployable. What a constructed corpus buys is control: because we generate the signature first and the surface form second, we can hold the attack fixed while varying the language, or hold the wording nearly fixed while inverting the security meaning. Those manipulations are what make the diagnostics possible, and they are very hard to obtain from scraped data.

2 Related Work

SMS spam and smishing detection. The SMS Spam Collection [1] established the standard formulation—binary classification over short English messages—and remains the most widely used benchmark. Later resources broadened coverage: SmishTank [16] contributes community-reported phishing SMS, and Salman et al. [14] assemble a large scam corpus and, most relevantly for us, show that detectors which perform well on clean text degrade sharply under character-, word- and sentence-level perturbation. Our work differs in the axis of stress: rather than perturbing text to preserve meaning while breaking features, we perturb text to *invert* meaning while preserving features.

Multilingual and code-mixed Indic NLP. MuRIL [9] and IndicNLP Suite [8] supply pre-trained encoders and benchmarks for Indian languages, including transliterated text; LaBSE [4] provides language-agnostic sentence embeddings that align translations in a shared space. These give the representational substrate a cross-lingually invariant signature model would need. To

our knowledge no existing scam-detection benchmark evaluates whether a detector assigns the *same structured attack label* to parallel realisations across Indian languages, which is the gap our CLSC measure targets.

Behavioural testing and contrast sets. CheckList [13] reframes NLP evaluation as behavioural testing with targeted capability probes. Contrast sets [5] perturb evaluation instances minimally so as to change the gold label, exposing decision boundaries that held-out accuracy hides. Our semantic-flip suite is a contrast set specialised to security semantics: the perturbation inverts the *action demanded of the recipient*, which is the field that determines whether a message is an attack at all.

Compositional generalisation. SCAN [11] and COGS [10] test whether models recombine familiar primitives in unfamiliar configurations. Scam taxonomies are naturally compositional—an actor, a pretext and a demanded action combine fairly freely—so the same question applies directly, and is operationally important because novel scams are typically recombinations rather than inventions.

Open-set recognition. Closed-set classifiers assign every input to a known class. Open-set recognition [15] and confidence-based out-of-distribution detection [7] instead permit abstention. Since attackers continually introduce new pretexts, a scam detector that cannot say “this does not match anything I know” will silently mislabel novel families.

LLMs and the evolution of social engineering. Hazell [6] documents how language models lower the cost of personalised spear phishing, and PEEK [2] uses them to study how phishing patterns evolve. If the surface form of attacks is now cheap to vary, evaluation that rewards surface pattern-matching becomes correspondingly less informative—which is the premise of this paper. Our tactic vocabulary draws on Cialdini’s persuasion principles [3].

3 The Attack Signature Framework

3.1 Definition

Let m be a message. Its **Attack Signature** is the tuple

$$S(m) = \langle a, p, t, i, c, g, T, E \rangle,$$

where a is the impersonated ACTOR, p the PRETEXT, t the TARGET asset, i the attacker INTENT, c the ACTION demanded of the recipient, g the kill-chain STAGE, $T \subseteq \mathcal{V}_{\text{tactic}}$ the set of persuasion TACTICS, and $E \subseteq \mathcal{V}_{\text{evidence}}$ the set of observable surface EVIDENCE cues. Each field is drawn from a closed controlled vocabulary (Table 1). The six single-valued fields are written F_1 , and the two set-valued fields F_M .

We call $\kappa(m) = \langle a, p, t, i, c \rangle$ the **signature core**: the fields that jointly determine whether, and how, a message is an attack.

The ACTION vocabulary includes the value **refrain**, which encodes an instruction *not* to perform an act. This single value is what lets the framework distinguish an attack from the advisory warning against it, and it is the mechanism the semantic-flip diagnostic probes.

Table 1: Controlled vocabularies. $|\mathcal{V}|$ excludes the **none** placeholder where present.

Field	$ \mathcal{V} $	Values (abbreviated)
ACTOR	10	bank, telecom, govt/police, courier, employer, e-commerce, known person, utility, tech support, investment platform
PRETEXT	12	account suspension, KYC expiry, parcel held, refund due, job offer, prize win, legal action, bill overdue, investment return, emergency help, routine notice, safety advisory
TARGET	7	OTP, card PAN, UPI PIN, account credentials, funds, identity document, device control
INTENT	6	credential theft, funds transfer, malware install, data harvest, advance fee, benign
ACTION	9	share code, click link, call number, install app, make payment, reply, scan QR, no action, <i>refrain</i>
STAGE	5	lure, engage, extract, launder, n/a
TACTIC	7	authority, urgency, fear, reward, social proof, reciprocity, familiarity
EVIDENCE	6	shortened URL, phone number, monetary amount, deadline, reference id, orthographic noise

3.2 Signature agreement

For two signatures S, S' define

$$\sigma(S, S') = \frac{|F_1| \cdot \frac{1}{|F_1|} \sum_{f \in F_1} \mathbb{I}[S_f = S'_f] + |F_M| \cdot \frac{1}{|F_M|} \sum_{f \in F_M} J(S_f, S'_f)}{|F_1| + |F_M|},$$

where J is Jaccard similarity and $\mathbb{I}$ the indicator. Thus $\sigma \in [0, 1]$, with $\sigma = 1$ iff the signatures are identical. Single-valued fields contribute 6/8 of the weight and set-valued fields 2/8.

3.3 Risk

Risk is a deterministic function of the signature, not a separately learned score:

$$\rho(S) = \begin{cases} 0 & \text{if } c = \text{refrain or } i = \text{benign,} \\ \min(1, 0.9 \lambda(i) + \beta(t)) & \text{otherwise,} \end{cases}$$

with λ an intent severity in $[0, 1]$ and β a small target-sensitivity bonus (both tabulated in the released code). Deriving risk from structure rather than fitting it directly is deliberate: it means a model can only change its risk estimate by changing its account of *what the attack is*, which is the behaviour we want to measure.

4 The BHARAT-SCAM-X Probe Corpus

4.1 Construction

We define twelve attack families as signature tuples instantiating documented Indian scam modalities: bank KYC/OTP theft, account-suspension OTP theft, KYC phishing links, “digital arrest” legal-threat extortion, courier-customs payment fraud, utility disconnection callbacks, task-based job fraud, advance-fee prize fraud, UPI refund PIN capture, remote-access tech support fraud, investment-return fraud, and relative-in-distress fraud.

Table 2: Corpus composition. 1,823 items, 1,563 unique strings; some items serve more than one diagnostic. No test item is string-identical to any training item.

Split	Suite	n	Purpose
train	in-distribution attacks	199	seen families, seen varieties
train	generic benign	84	class balance
train	hard negatives	13	benign, scam vocabulary
train	supervised flip pairs	60	makes refrain learnable
test	in-distribution	71	held-out paraphrases
test	cross-lingual	506	six unseen varieties, zero-shot
test	semantic flips	293	143 minimal-edit contrast pairs
test	compositional	72	unseen component recombinations
test	open-set	171	two entirely unseen families
test	parallel (CLSC)	90	one signature $\times$ 9 varieties
test	hard negatives	55	benign, scam vocabulary
test	generic benign	212	ordinary traffic

Surface realisation is compositional. For each language variety we define independent lexical inventories for actor prefixes, pretext clauses, action clauses (keyed by the $\langle \text{ACTION}, \text{TARGET} \rangle$ pair), urgency intensifiers, and *refrain* clauses. A message is assembled by selecting one realisation per slot under a seeded RNG. Because the signature is chosen first, every item carries a gold label by construction rather than by annotation, and inter-annotator agreement is not a confound.

Language varieties. Nine: English (EN); Hindi in Devanagari (HI) and romanised (HI_LATN); Telugu in Telugu script (TE) and romanised (TE_LATN); Tamil (TA); Kannada (KN); and English–Hindi (CM_HI) and English–Telugu (CM_TE) code-mixed varieties. Three varieties (EN, HI, HI_LATN) are seen in training; the remaining six are held out entirely, so cross-lingual evaluation is zero-shot.

4.2 Constructing minimal-edit semantic flips

For an attack message realised as $[prefix] [pretext] [action]$, the corresponding flip is generated with the *same RNG seed*, so the prefix and pretext clauses are string-identical, and only the action clause is replaced by its refrain counterpart. The flip’s gold signature retains the actor but sets $p = \text{**safety_advisory**}$, $i = \text{**benign**}$, $c = \text{**refrain**}$, hence $\rho = 0$.

This construction is the paper’s central instrument, so its minimality is measured rather than asserted: across the 143 held-out pairs, mean token-level Jaccard overlap is **0.552**. An example pair:

Attack “Dear customer, Your KYC has expired. Share the OTP sent to your phone.”
($\rho = 1.0$)

Flip “Dear customer, Your KYC has expired. Never share the OTP sent to your phone with anyone.” ($\rho = 0.0$)

4.3 Splits and diagnostic suites

Table 2 gives the composition. Three design choices deserve comment.

Flip supervision. Flips for five of the ten known families are placed in *training*. Without this, no model could ever predict **refrain**, and failure on the flip suite would be a tautology rather than a finding. The held-out flip suite therefore measures whether the attack/advisory distinction *transfers* to unseen families and varieties.

Class balance. Training is 229 attack / 127 benign. An earlier version of this corpus was 94% attacks, under which every trained baseline collapsed to predicting the majority class and scored near-perfectly on all-positive suites. We report this because it is a trap the task invites.

Mixed evaluation conditions. Each reported condition in Table 3 contains both attack and benign items, so macro- F_1 is meaningful and a degenerate all-positive predictor cannot score well.

5 Diagnostics

Cross-Lingual Signature Consistency (CLSC). For a parallel group G containing one realisation of a single signature per language variety, $\text{CLSC}(G) = \binom{|G|}{2}^{-1} \sum_{(u,v)} \sigma(\hat{S}(u), \hat{S}(v))$, averaged over groups. This measures *self-consistency*: whether the model tells the same story about the same attack regardless of how it is written. It is independent of whether that story is correct, which we measure separately as agreement with gold.

Semantic Flip Sensitivity (SFS). Over contrast pairs (m^+, m^-) we report the *flip-success rate* $\Pr[\hat{\rho}(m^+) - \hat{\rho}(m^-) \geq 0.5]$, the mean risk drop, and the mean token overlap of the pair. The threshold 0.5 encodes the requirement that a security-meaning inversion move risk across the decision midpoint, not merely nudge it.

Compositional signature recovery. We construct four probes whose $\langle \text{PRETEXT}, \text{ACTION} \rangle$ pairings never co-occur in training although both constituents appear separately—e.g. a bank KYC pretext with a call-number action. We report exact signature-core accuracy and mean σ against gold.

Open-set behaviour. Two families are withheld entirely. We score novelty by maximum softmax probability over a family classifier [7] and report AUROC for separating held-out-family items from known-family items.

6 Baselines

Keyword-Rule flags a message when at least two of 30 hand-written scam-indicative patterns match (covering Latin, Devanagari, Telugu, Tamil and Kannada cues). **TFIDF-Word-LR** is word 1–2-gram TF-IDF with logistic regression. **TFIDF-Char-LR** is character 2–5-gram TF-IDF (`char_wb`) with logistic regression. **Struct-Sig-Char** is our structured model: one logistic-regression head per single-valued signature field over the same character features, with risk computed from the predicted tuple by ρ . All models see identical training data; the seed is fixed at 20260920.

We note explicitly that no pre-trained multilingual encoder baseline is included. MuRIL [9] or LaBSE [4] are the natural substrates for a cross-lingually invariant signature model, and we expect them to materially improve the cross-lingual numbers in particular. Their absence is a limitation of this study, not a claim about their performance; the released harness is written so that adding an encoder requires replacing one feature function.

7 Results

In-distribution accuracy is close to uninformative. Every trained baseline exceeds 0.91 macro- F_1 in-distribution (Table 3), and the structured model is perfect. Ranking systems on this column would put TFIDF-Char-LR and Struct-Sig-Char within 0.06 of each other. Their

Table 3: Binary detection, macro- F_1 . Every condition mixes attack and benign items; condition sizes are given as (+pos/−neg).

	IID (+71/−33)	Cross-ling (+506/−234)	Sem. flip (+143/−150)	Composit. (+72/−267)	Open-set (+171/−267)
Keyword-Rule	0.706	0.686	0.570	0.747	0.640
TFIDF-Word-LR	0.917	0.943	0.677	0.891	0.946
TFIDF-Char-LR	0.942	0.936	0.700	0.920	0.965
Struct-Sig-Char	1.000	0.888	0.781	0.918	0.920

Table 4: False-positive rate on benign text (lower is better). “Semantic flips” are genuine security advisories.

	Hard negatives	Semantic flips	Generic benign
Keyword-Rule	0.200	0.420	0.000
TFIDF-Word-LR	0.382	0.560	0.009
TFIDF-Char-LR	0.273	0.507	0.000
Struct-Sig-Char	0.000	0.293	0.000

flip-success rates differ by a factor of 3.2 (Table 5). The in-distribution column simply does not carry the information a deployment decision needs.

Security-meaning inversion defeats lexical models. On pairs sharing 55.2% of their tokens, TFIDF-Char-LR achieves 0.175 flip-success, and the keyword rule 0.028—at or below what a coin flip on the pair would yield. Table 4 states the operational consequence: the word-level model flags 56.0% of genuine “never share your OTP” advisories as attacks. A filter with that property would suppress precisely the messages banks send to reduce fraud.

Structure helps, and does not suffice. Struct-Sig-Char improves flip-success to 0.559 and drives hard-negative FPR to zero. This supports the paper’s thesis: a model forced to name the demanded action can represent the attack/advisory distinction, and this transfers to families whose flips it never saw. But 0.559 means it still misreads 44% of inverted messages, and it is the *worst* model cross-lingually (0.888 vs. 0.943). The reason is mechanical and worth stating plainly: predicting six fields correctly is strictly harder than predicting one bit, so structure converts a binary problem into six opportunities to fail. Structured prediction is not free.

Recombination is where it breaks. Table 6 contains the study’s sharpest result. On compositional probes—built *entirely* from components the model saw in training, merely combined differently—exact signature-core recovery falls from 1.000 to **0.056**. Per-field accuracy degrades gracefully (ACTION 0.681, INTENT 0.611), but the conjunction almost never survives. Notably, binary detection on this same suite stays high (0.918, Table 3): the model still says “attack” while being wrong about essentially every particular of what the attack is. Since real novel scams are overwhelmingly recombinations of familiar actors, pretexts and asks, this is the failure mode with the most direct operational cost, and it is invisible to binary evaluation.

Cross-lingual consistency degrades by script distance. CLSC is 1.000 on the three training varieties and 0.787 on the six held-out ones (0.837 overall). Agreement with gold, by variety, tracks script and lexical distance from training: CM_HI 0.750, CM_TE 0.738, KN 0.625, TA 0.613, TE_LATN 0.475, TE 0.450. Code-mixed romanised text transfers best—it shares English tokens—while Telugu in native script transfers worst. This is the expected behaviour of

Table 5: Semantic Flip Sensitivity over 143 minimal-edit contrast pairs.

	Flip-success	Mean risk drop	Token overlap
Keyword-Rule	0.028	0.063	0.552
TFIDF-Word-LR	0.189	0.256	0.552
TFIDF-Char-LR	0.175	0.277	0.552
Struct-Sig-Char	0.559	0.494	0.552

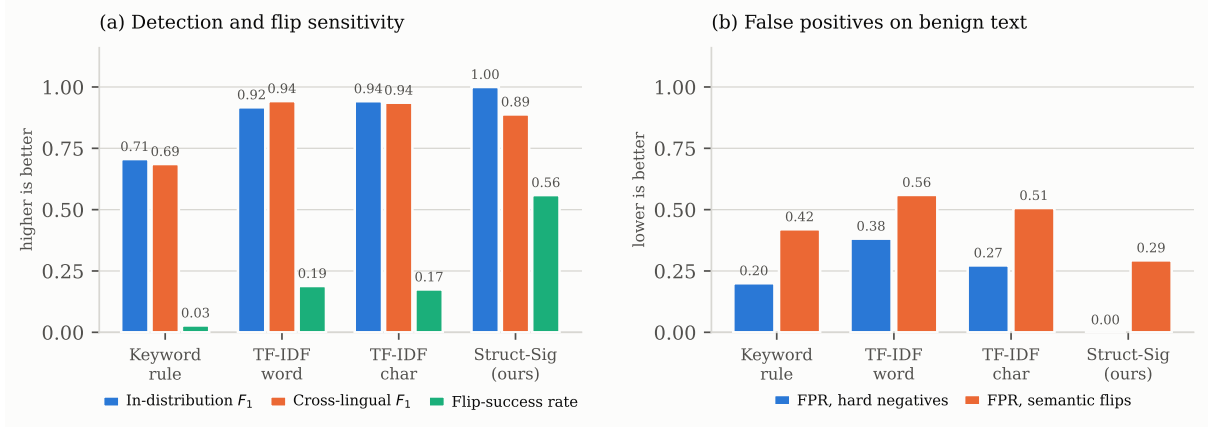


Figure 1: The evaluation gap. (a) In-distribution F_1 is high for every trained baseline, but flip-success—the ability to recognise that inverting one clause inverts the security meaning—is far lower. (b) All baselines fire on a substantial fraction of genuine security advisories; the structured model fires least.

character n -grams and is precisely what a shared multilingual embedding space should repair; see Section 9.

Open-set numbers here are optimistic. Maximum-softmax-probability novelty detection reaches AUROC 0.905. We caution against reading this as encouraging. Our two held-out families introduce actor and pretext vocabulary absent from training, so their novelty is lexically visible; a genuinely hard open-set case is a new pretext expressed in familiar vocabulary. We report the number for completeness and regard the suite as the weakest of the five.

8 Discussion

Three observations generalise beyond our corpus.

The metric you optimise determines the failure you ship. The in-distribution column of Table 3 and the flip column of Table 5 rank models differently. A team that reports only the former can improve its headline number while making its product worse at the thing users care about.

False positives on advisories are a security harm, not just an annoyance. Every baseline flags between 29% and 56% of genuine “never share your OTP” messages. Anti-fraud advisories are a primary defence channel; a filter that suppresses them degrades the defence it is meant to support. We suggest advisory FPR be reported as standard in this literature.

Structured prediction trades breadth for depth. Our structured model is best where semantics matter (flips, hard negatives) and worst where surface transfer matters (cross-lingual). This is not an argument against structure; it is an argument for pairing structure with a representation that is already cross-lingually aligned.

Table 6: Attack Signature recovery, Struct-Sig-Char. “Exact core” requires all five core fields correct.

Suite	n	Exact core	Mean σ	actor	intent	action
In-distribution	71	1.000	0.750	1.000	1.000	1.000
Cross-lingual	506	0.642	0.600	0.792	0.765	0.919
Compositional	72	0.056	0.441	0.431	0.611	0.681
Open-set	171	0.000	0.342	0.000	0.468	0.959

9 Limitations

The corpus is synthetic. Every message is template-composed. Real scam SMS contain typos, unusual spacing, emoji, URL obfuscation, sender-ID spoofing and length constraints we do not model, and real benign traffic is far more diverse than ours. Our results characterise model behaviour on controlled contrasts; they do not estimate field performance, and we would regard any deployment claim based on them as unsupported.

Template artifacts may be learnable. Because attacks and benign items are generated by different procedures, a model could in principle exploit generation artifacts rather than meaning. We mitigate this by composing benign messages with the same actor-prefix envelopes as attacks, and by making flip pairs share prefix and pretext exactly; but we cannot rule it out. This is the strongest reason to treat the corpus as diagnostic rather than as a leaderboard.

No neural or LLM baselines. We evaluate feature-based models only. Pre-trained multi-lingual encoders and instruction-tuned LLMs will behave differently—plausibly much better on cross-lingual transfer, and, we would predict, still imperfectly on flips. Establishing that is the obvious next step.

Scale and coverage. Twelve families, nine varieties, 1,823 items. Major Indian languages including Bengali, Marathi, Gujarati, Malayalam, Odia and Punjabi are absent. Set-valued fields (TACTIC, EVIDENCE) are not predicted by our structured baseline at all, which is why its mean σ is capped near 0.75 even at perfect core accuracy.

Single annotator of record. Vocabularies and non-English realisations were authored by the present author without independent linguistic review. Native-speaker validation of the Telugu, Tamil and Kannada material is needed before the corpus is used as a shared benchmark.

The risk function is stipulated. ρ is hand-specified rather than calibrated against harm data, so absolute risk values carry no external meaning; only relative comparisons within our evaluation are interpretable.

10 Ethical Considerations

This work releases synthetic scam messages. We judge the marginal uplift to attackers to be negligible: the messages are template-generated instances of publicly documented fraud patterns, shorter and less persuasive than material already circulating, and contain no real institution branding, no working URLs (all links are non-resolving placeholders), no real phone numbers, and no personal data. The defensive value of a corpus that lets researchers measure advisory false-positive rates appears clearly greater.

We flag one asymmetric risk. A detector tuned on this corpus could be deployed and, through the compositional and cross-lingual failures we document, silently under-protect speakers of languages not represented in its training data. Users of the corpus should treat Table 6 as a warning rather than a scoreboard.

11 Conclusion

Scam detection is usually evaluated as though the problem were recognising scam-like text. We have argued that the problem is representing an attack, and that the two come apart measurably. A model at 0.942 in-distribution macro- F_1 recovers the right attack structure for 5.6% of recombined attacks and treats half of all genuine security advisories as attacks. Making structure explicit helps materially—tripling flip sensitivity and eliminating hard-negative false positives—but leaves most of the gap open, and costs cross-lingual robustness. We release the schema, the corpus and the harness so that the remaining gap can be measured rather than assumed away.

Availability

The Attack Signature schema, corpus generator, evaluation harness and all numbers reported here are reproducible from a single fixed seed (20260920). The corpus is regenerated by `python3 src/generate.py` and all tables by `python3 src/evaluate.py`.

Acknowledgements

Portions of the software implementation and the drafting of this manuscript were carried out with the assistance of an AI coding assistant. All design decisions, the framework, the experimental protocol and the claims made here are the author’s, who takes full responsibility for the content.

References

- [1] Tiago A. Almeida, José María Gómez Hidalgo, and Akebo Yamakami. Contributions to the study of SMS spam filtering: New collection and results. In *Proceedings of the 11th ACM Symposium on Document Engineering (DocEng)*, pages 259–262, 2011.
- [2] Fengchao Chen, Tingmin Wu, Van Nguyen, Shuo Wang, Alsharif Abuadbba, and Carsten Rudolph. PEEK: Phishing evolution framework for phishing generation and evolving pattern analysis using large language models. *arXiv preprint arXiv:2411.11389*, 2024.
- [3] Robert B. Cialdini. *Influence: The Psychology of Persuasion*. Collins, revised edition, 2007.
- [4] Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pages 878–891, 2022.
- [5] Matt Gardner, Yoav Artzi, Victoria Basmov, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, Nitish Gupta, Hananeh Hajishirzi, Gabriel Ilharco, Daniel Khoshnab, Kevin Lin, Jiangming Liu, Nelson F. Liu, Phoebe Mulcaire, Qiang Ning, Sameer Singh, Noah A. Smith, Sanjay Subramanian, Reut Tsarfaty, Eric Wallace, Ally Zhang, and Ben Zhou. Evaluating models’ local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1307–1323, 2020.
- [6] Julian Hazell. Spear phishing with large language models. *arXiv preprint arXiv:2305.06972*, 2023.
- [7] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.

- [8] Divyanshu Kakwani, Anoop Kunchukuttan, Satish Golla, Gokul N.C., Avik Bhattacharyya, Mitesh M. Khapra, and Pratyush Kumar. IndicNLPSuite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for indian languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961, 2020.
- [9] Simran Khanuja, Diksha Bansal, Sarvesh Mehtani, Savya Khosla, Atreyee Dey, Balaji Gopalan, Dilip Kumar Margam, Pooja Aggarwal, Rajiv Teja Nagipogu, Shachi Dave, Shruti Gupta, Subhash Chandra Bose Gali, Vish Subramanian, and Partha Talukdar. MuRIL: Multilingual representations for indian languages. *arXiv preprint arXiv:2103.10730*, 2021.
- [10] Najoung Kim and Tal Linzen. COGS: A compositional generalization challenge based on semantic interpretation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9087–9105, 2020.
- [11] Brenden M. Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 2873–2882, 2018.
- [12] Ministry of Home Affairs, Government of India. Cyber crime statistics for 2025 reported to parliament, 2026. Figures tabled February 2026: 28.15 lakh complaints and Rs.22,495 crore in reported losses for calendar year 2025.
- [13] Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912, 2020.
- [14] Muhammad Salman, Muhammad Ikram, and Mohamed Ali Kaafar. An empirical analysis of SMS scam detection systems. *arXiv preprint arXiv:2210.10451*, 2022.
- [15] Walter J. Scheirer, Anderson de Rezende Rocha, Archana Sapkota, and Terrance E. Boult. Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7):1757–1772, 2013.
- [16] Daniel Timko and Muhammad Lutfor Rahman. Smishing dataset I: Phishing SMS dataset from smishtank.com, 2024. Community-contributed smishing corpus.